Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/15/2023
PI Name:	Paul, Amber Ph.D.		
Project Title:	Altered Gravity during Space Flight contributes to Immune Dysfunction through Stress-induced Heat Shock Proteins		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 236-5		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	Updated contact information. PI moved to Embry-Riddle Aeronautical University after research project ended (Ed., 2/15/23)		
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	07/31/2017	**End Date:**	07/23/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
Contact Email:	npphelp@usra.edu		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/23/2021 per PI (Ed., 1/29/21) NOTE: Extended to 10/31/2020 per PI (Ed., 11/13/19) NOTE: Extended to 7/30/2019 per T. Simpson/USRA NPP (Ed., 5/6/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (MENTOR/ NASA Ames Research Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In space, astronauts are more susceptible to pathogens, viral reactivation, and immune dysfunction, which pose limits to their health and the mission. Interestingly, during space flight stress-inducible heat shock proteins (Hsp) are robustly induced, while overexpression of Hsp has been implicated in immune dysregulation; therefore, Hsp may be critically involved in regulating immunity during space flight. To test this, wildtype and Hsp70 knockout Drosophila melanogaster will be exposed to ground-based hypergravity conditions, followed by quantitative PCR analysis of immune gene expression. In addition, human U937 and HL-60 cells will be transfected with small interfering RNA targeting Hsp70 or an Hsp70 inhibitor, and exposed to simulated microgravity conditions using a NASA developed high-aspect rotating wall vessel. In brief, a critical need to evaluate the relationship between Hsp and immunity during space flight is necessary. Since space travel may become available to the general public in the not too distant future, and for the possibility of long-term space missions, a more comprehensive evaluation of the molecules responsible for immune dysregulation observed during space flight is required.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research in the field of space immunology is indispensable to understand human physiology in space and warrants critical evaluation by NASA for successful crewed missions to the Moon and Mars. Multiple reports identified a role for increased Hsp in response to space flight-induced stress; while Hsp are typically transient and short-lived, a comprehensive understanding of how excess or prolonged Hsp contributes to the development of immune dysregulation is required. This research is not only relevant to space biology, but is for life and health sciences as well, in particular in immune mechanisms relevant in tumor biology and during immune cell dysregulation in cancer development.
Task Progress:	All fly studies have been completed. Continued translational work on determining the mechanisms between the stress response as it relates to heat shock protein 70 (Hsp70), and its contribution to immunity, in the mammalian system are ongoing. Differential populations percentages have been identified and reported in simulated microgravity models of hindlimb unloading in mice and high-aspect rotating wall bioreactors with human white blood cell cultures. The connection between immune population differentials and expression of Hsp70 are under investigation in the context of simulated microgravity. Further, Hsp70 inhibition studies in monocytic cells display changes to immune function, both of which are being further validated with manuscript preparations underway at Embry-Riddle Aeronautical University (ERAU).
Bibliography Type:	Description: (Last Updated: 04/17/2023)
Abstracts for Journals and Proceedings	Paul AM, Cheng-Campbell M, Blaber EA, Anand S, Bhattacharya S, Zwart SR, Crucian BE, Smith SM, Meller R, Grabham P, Beheshti A. "Beyond low-Earth orbit: Characterizing immune and microRNA differentials following simulated deep spaceflight conditions in Mice." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Articles in Peer-reviewed Journals	Rubinstein L, Paul AM, Houseman C, Abegaz M, Ruiz ST, O'neil N, Kunis G, Ofir R, Cohen J, Ronca AE, Globus RK, Tahimic CGT. "Placenta-expanded stromal cell therapy in a rodent model of simulated weightlessness. " Cells. 2021 Apr 19;10(4):940. https://doi.org/10.3390/cells10040940 ; PMID: 33921854; PMCID: PMC8073415 , Apr-2021
Articles in Peer-reviewed Journals	Paul AM, Cheng-Campbell M, Blaber EA, Anand S, Bhattacharya S, Zwart SR, Crucian BE, Smith SM, Meller R, Grabham P, Beheshti A. "Immune and microRNA differentials post-simulated deep spaceflight." Immunol. 2021 May 1;206(1_Supplement):11.03. https://doi.org/10.4049/jimmunol.206.Supp.11.03 ; PMID: 33376970; PMCID: PMC7756144 , May-2021
Articles in Peer-reviewed Journals	Paul AM, Overbey EG, da Silveira WA, Szewczyk N, Nishiyama NC, Pecaut MJ, Anand S, Galazka JM, Mao XW. "Immunological and hematological outcomes following protracted low dose/low dose rate ionizing radiation and simulated microgravity." Sci Rep. 2021 Jun 1;11(1):11452. https://doi.org/10.1038/s41598-021-90439-5 ; PMID: 34075076; PMCID: PMC8169688 , Jun-2021
Articles in Peer-reviewed Journals	Mhatre SD, Iyer J, Petereit J, Dolling-Boreham RM, Tyryshkina A, Paul AM, Gilbert R, Jensen M, Woolsey RJ, Anand S, Sowa MB, Quilici DR, Costes SV, Girirajan S, Bhattacharya S. "Artificial gravity partially protects space-induced neurological deficits in Drosophila melanogaster." Cell Rep. 2022 Sep 6;40(10):111279. https://doi.org/10.1016/j.celrep.2022.111279 ; PMID: 36070701 , Sep-2022